FINAL CAPSTONE PROJECT

Project Name: SafeCryptoStocks

Deployed by: Abhirami Kannan

Cloud Platform: Microsoft Azure

Infrastructure as a Code: Terraform

1. **Github project** used: https://github.com/Abhiramikannan/Azure-Final-Capstone-Project.git
2. Created a vm and installed terraform using the official website
   a. https://developer.hashicorp.com/terraform/install#linux
   b. Installed az client
   c. curl -sL https://aka.ms/InstallAzureCLIDeb | sudo bash
   d. az login

➢ pushed the **terraform code** to github ,also created a .gitignore to ignore some files ,so that I wont get error of secrets

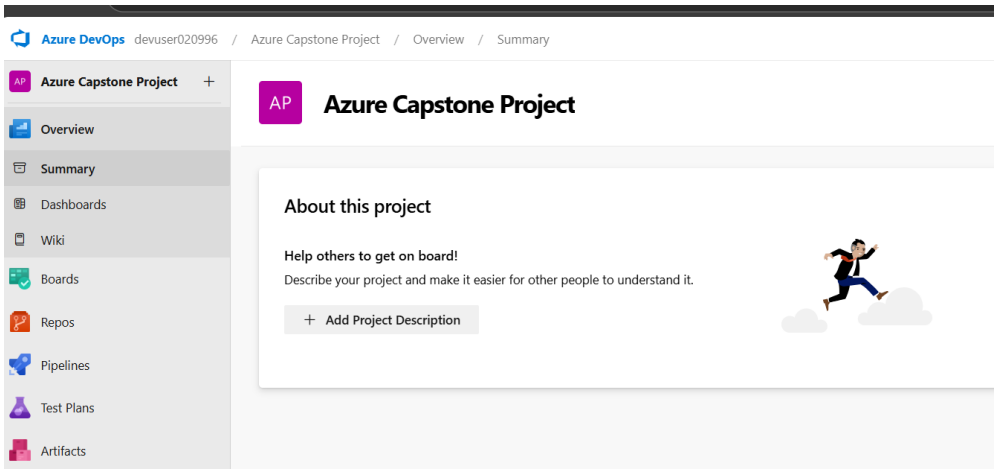
```
root@abhi-vm:/home/abhi/Azure-Final-Capstone-Project# cat .gitignore
terraform-code/.terraform/
terraform-code/*.tfstate
terraform-code/*.tfstate.*
terraform-code/.terraform.lock.hcl
*.tfplan
*.backup
```

1. install docker using official documentation:

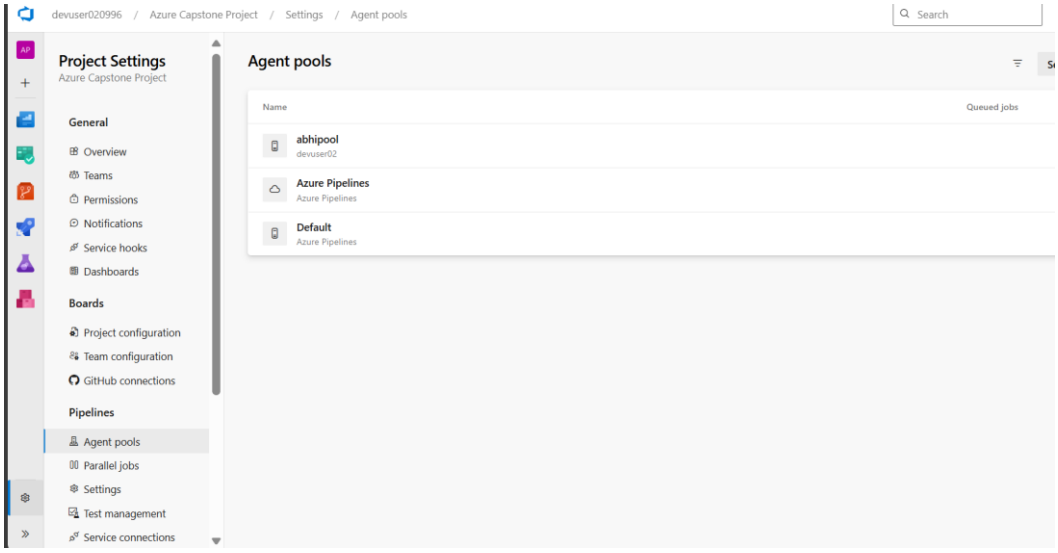
i) Give docker permissions to the user abhi:

ii) sudo usermod -aG docker abhi

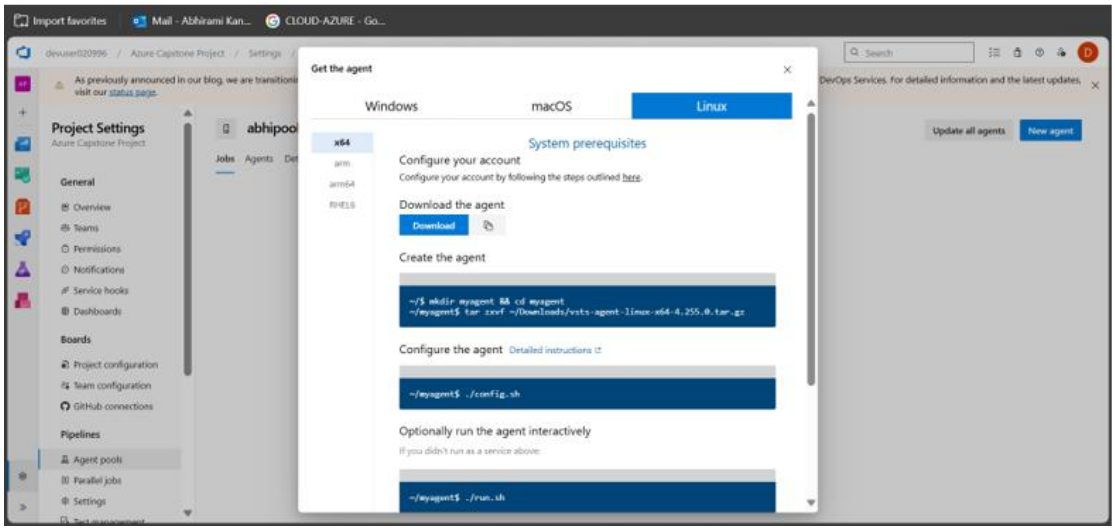
3. created a project in azure



4. created an agentpool
    a. projectsettings-> agent pools -> add pool



    b. install the agent in the virtual machine using the following commands:




```
abhi@abhi-vm:~$ wget https://download.agent.dev.azure.com/agent/4.255.0/vsts-agent-linux-x64-4.255.0.tar.gz
--2025-06-19 06:57:14--  https://download.agent.dev.azure.com/agent/4.255.0/vsts-agent-linux-x64-4.255.0.tar.gz
Resolving download.agent.dev.azure.com (download.agent.dev.azure.com)... 23.215.0.43, 23.215.0.37, 2600:1408:ec00:1f::1735:23d1, ...
Connecting to download.agent.dev.azure.com (download.agent.dev.azure.com)|23.215.0.43|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 147552820 (141M) [application/octet-stream]
Saving to: 'vsts-agent-linux-x64-4.255.0.tar.gz'

vsts-agent-linux-x64-4.255.0.tar.gz                 100%[=================================================================================
===>] 140.72M   246MB/s    in 0.6s

2025-06-19 06:57:14 (246 MB/s) - 'vsts-agent-linux-x64-4.255.0.tar.gz' saved [147552820/147552820]

abhi@abhi-vm:~$
```

```
~/$ mkdir myagent && cd myagent
~/myagent$ tar zxvf ~/Downloads/vsts-agent-linux-x64-4.255.0.tar.gz
```

Changed the location tar zxvf ~/Downloads/vsts-agent....to

tar zxvf ../vsts-agent-linux.....

- Because my file is in that location not in Downloads folder

```
abhi@abhi-vm:~/myagent$ ./config.sh

  agent v4.255.0                    (commit 470b366)

>> End User License Agreements:

Building sources from a TFVC repository requires accepting the Team Explorer Everywhere End User License Agreement. This step is not required for building sources from G
t repositories.

A copy of the Team Explorer Everywhere license agreement can be found at:
  /home/abhi/myagent/license.html

Enter (Y/N) Accept the Team Explorer Everywhere license agreement now? (press enter for N) >
```

- Generate pat:
- Azure devops->user settings->**pat**->full access->copy pat
- **Server url is**: https://dev.azure.com/devuser020996/

```
>> End User License Agreements:

Building sources from a TFVC repository requires accepting the Team Explorer Everywhere End User License Agreement. This step is not required for building sources from Gi
t repositories.

A copy of the Team Explorer Everywhere license agreement can be found at:
  /home/abhi/myagent/license.html

Enter (Y/N) Accept the Team Explorer Everywhere license agreement now? (press enter for N) > y

>> Connect:

Enter server URL > https://dev.azure.com/devuser020996/
Enter authentication type (press enter for PAT) >
Enter personal access token > ************************************************************************************
Connecting to server ...

>> Register Agent:

Enter agent pool (press enter for default) >
```

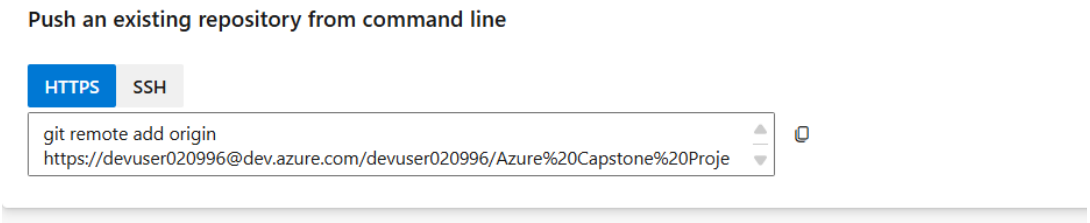
```
>> Register Agent:

Enter agent pool (press enter for default) > abhipool
Enter agent name (press enter for abhi-vm) >
Scanning for tool capabilities.
Connecting to the server.
Successfully added the agent
Testing agent connection.
Enter work folder (press enter for _work) >
025-06-19 07:12:19Z: Settings Saved.
abhi@abhi-vm:~/myagent$
```

Run the agent:

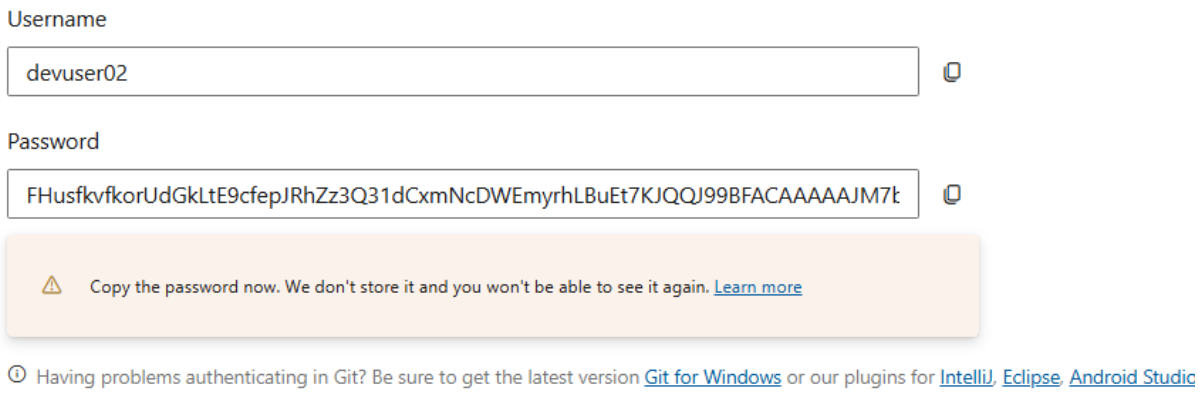
```
abhi@abhi-vm:~/myagent$ ./run.sh
Scanning for tool capabilities.
Connecting to the server.
2025-06-19 07:14:31Z: Listening for Jobs
```

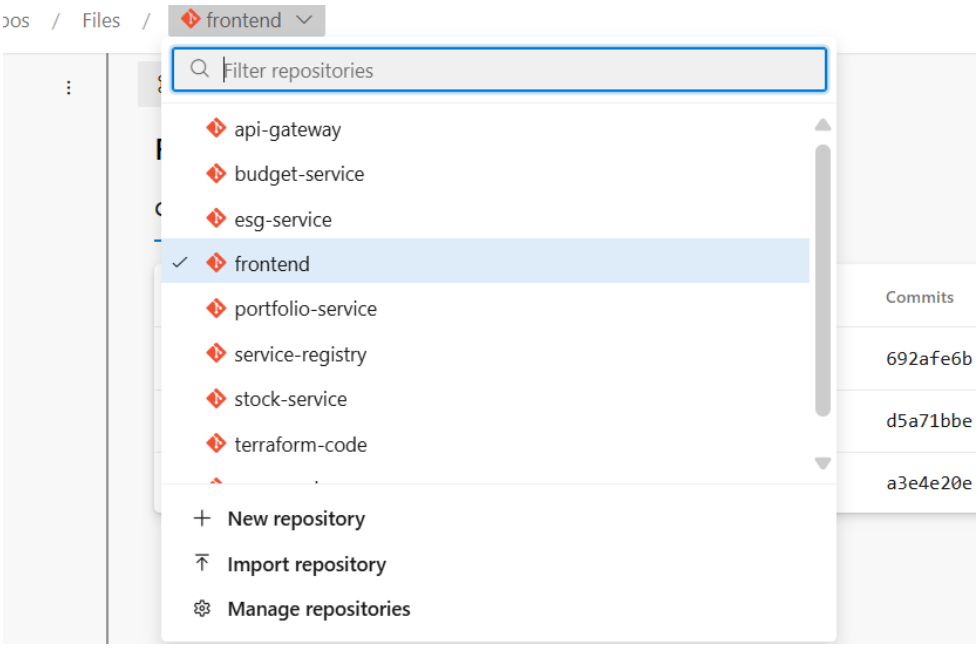
5. Splitted the services as the different repository in azure devops

i. Moved each folder into root and done the operations of git init,git commit and used the command which we get from azure devops while creating a repository and pushed using credentials in azure devops for that repository and push succeeded.

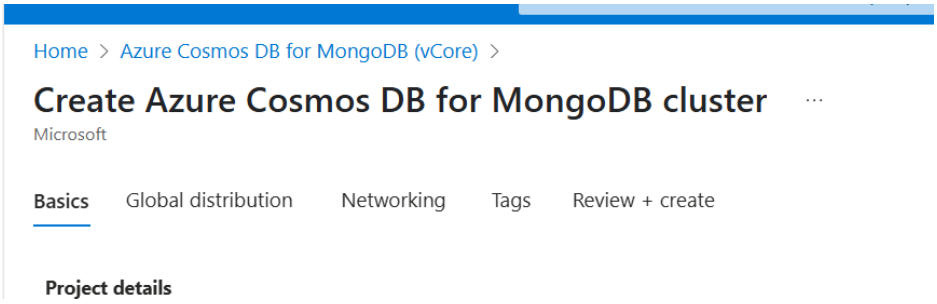


Credentials:

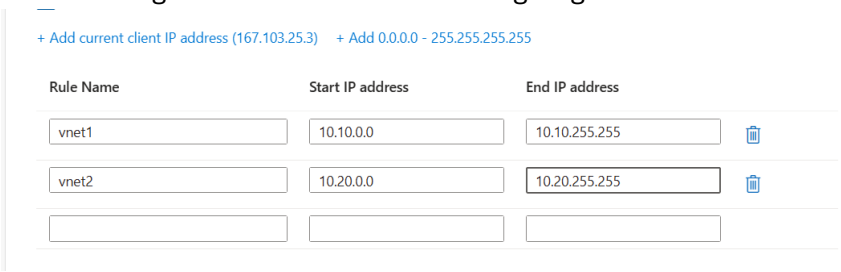


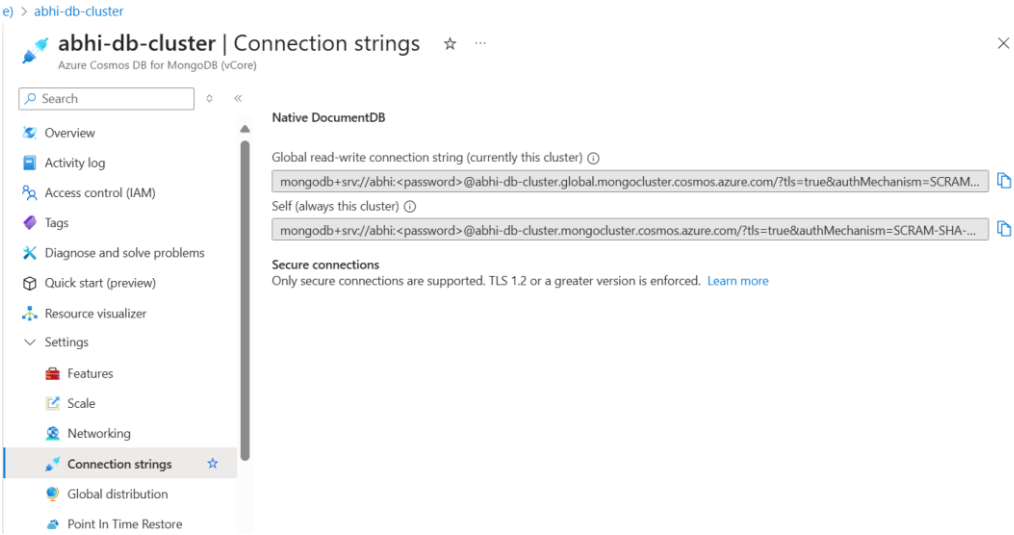


6. Created the database:



Address range of vnet1 and vnet2 we are giving:





7. I created a **service principal** for the non azure resources like terraform to access azure resources(required for key vault access policy in terraform)
8. **Storage Blob Data Contributor:** This role I have given to service principal
    - assigned assign the role that allows your Service Principal to read and write blobs (your Terraform state file) in that storage account.

```
devuser [ ~ ]$ az storage account show --name abhicapstonestorage --resource-group abhi-resource-group --query id --output tsv
/subscriptions/1674f375-e996-4423-bd25-e0e8f6e76d13/resourceGroups/abhi-resource-group/providers/Microsoft.Storage/storageAccounts/abhicapstonestorage
devuser [ ~ ]$ az role assignment create \
  --role "Storage Blob Data Contributor" \
  --assignee "df6ba479-db72-463c-bef0-59a76e3b6bf4" \
  --scope "$STORAGE_ACCOUNT_ID"
{
  "condition": null,
  "conditionVersion": null,
  "createdBy": null,
  "createdOn": "2025-06-21T05:26:24.769180+00:00",
  "delegatedManagedIdentityResourceId": null,
  "description": null,
  "id": "/subscriptions/1674f375-e996-4423-bd25-e0e8f6e76d13/resourceGroups/abhi-resource-group/providers/Microsoft.Storage/storageAccounts/abhicapstonestorage/providers/
Microsoft.Authorization/roleAssignments/82bbdee1-68f7-4467-bdef-9a6ebe099a28",
  "name": "82bbdee1-68f7-4467-bdef-9a6ebe099a28",
  "principalId": "1c6ad34d-73d1-4c1a-b38d-735be951325d",
  "principalType": "ServicePrincipal",
  "resourceGroup": "abhi-resource-group",
  "roleDefinitionId": "/subscriptions/1674f375-e996-4423-bd25-e0e8f6e76d13/providers/Microsoft.Authorization/roleDefinitions/ba92f5b4-2d11-453d-a403-e96b0029c9fe",
  "scope": "/subscriptions/1674f375-e996-4423-bd25-e0e8f6e76d13/resourceGroups/abhi-resource-group/providers/Microsoft.Storage/storageAccounts/abhicapstonestorage",
  "type": "Microsoft.Authorization/roleAssignments",
  "updatedBy": "3cb026dd-3f46-470c-b65c-e0bd92d4fd19",
  "updatedOn": "2025-06-21T05:26:25.174186+00:00"
}
devuser [ ~ ]$
```

Run these two commands sequentially in your terminal on the VM. After they complete, your Service Principal will have the necessary permissions on your Terraform state storage account

9. Create service connection for the service principal u explicitly created
10. Then created a pipeline for my terraform code and deployed it in both the clusters
11. **The infrastructure includes**:
    a. Created the backend configurations to access the tfstatefile in the remote ...so this will compare the .tf file and tfstate in the remote .If state files are same then it will say no changes otherwise apply changes

```
  # THIS IS THE CORRECT PLACE FOR THE BACKEND BLOCK
  backend "azurerm" {
    resource_group_name  = "abhi-resource-group"
    storage_account_name = "abhicapstonestorage"
    container_name       = "tfstate"
    key                  = "terraform.tfstate"
  }
}
```

    b. Created the resource group,Acr

```
resource "azurerm_container_registry" "acr" {
  name                = "abhiacr1"
  resource_group_name = azurerm_resource_group.myrg.name
  location            = azurerm_resource_group.myrg.location
  sku                 = "Standard"
  admin_enabled       = true # Enable admin user for push/pull (consider service principals for prod)

  identity {
    type = "SystemAssigned" # ACR's identity can be SystemAssigned for its own operations
  }

  depends_on = [azurerm_resource_group.myrg]
}
```

- Type=systemassigned so Acr can securely talk to other azure services
- **Allow ACR to authenticate to other Azure services without storing secrets.**
- Best practice and passwordless authentication for future integrations

c. Vnets and subnets

```
# ==========================================================================================
# Virtual Networks and Subnets
# ==========================================================================================

# VNet 1 in Central India (Primary VNet for AKS)
resource "azurerm_virtual_network" "vnet1" {
  name                = "abhi-vnet1"
  address_space       = ["10.10.0.0/16"]
  location            = var.vnet1_location
  resource_group_name = var.resource_group_name

  depends_on = [azurerm_resource_group.myrg]
}

resource "azurerm_subnet" "vnet1_subnet1" {
  name                 = "abhi-vnet1-subnet1" # Subnet for AKS default node pool
  resource_group_name  = var.resource_group_name
  virtual_network_name = azurerm_virtual_network.vnet1.name
  address_prefixes     = ["10.10.1.0/24"]

  depends_on = [azurerm_virtual_network.vnet1]
}

resource "azurerm_subnet" "vnet1_subnet2" {
  name                 = "abhi-vnet1-subnet2" # Subnet for AKS user node pool
  resource_group_name  = var.resource_group_name
  virtual_network_name = azurerm_virtual_network.vnet1.name
  address_prefixes     = ["10.10.2.0/24"]

  depends_on = [azurerm_virtual_network.vnet1]
}
```

```
# VNet 2 in West Europe (Secondary VNet for AKS)
resource "azurerm_virtual_network" "vnet2" {
  name                = "abhi-vnet2"
  address_space       = ["10.20.0.0/16"]
  location            = var.vnet2_location
  resource_group_name = var.resource_group_name

  depends_on = [azurerm_resource_group.myrg]
}

resource "azurerm_subnet" "vnet2_subnet1" {
  name                 = "abhi-vnet2-subnet1" # Subnet for AKS default node pool
  resource_group_name  = var.resource_group_name
  virtual_network_name = azurerm_virtual_network.vnet2.name
  address_prefixes     = ["10.20.1.0/24"]

  depends_on = [azurerm_virtual_network.vnet2]
}

resource "azurerm_subnet" "vnet2_subnet2" {
  name                 = "abhi-vnet2-subnet2" # Subnet for AKS user node pool
  resource_group_name  = var.resource_group_name
  virtual_network_name = azurerm_virtual_network.vnet2.name
  address_prefixes     = ["10.20.2.0/24"]

  depends_on = [azurerm_virtual_network.vnet2]
}
```

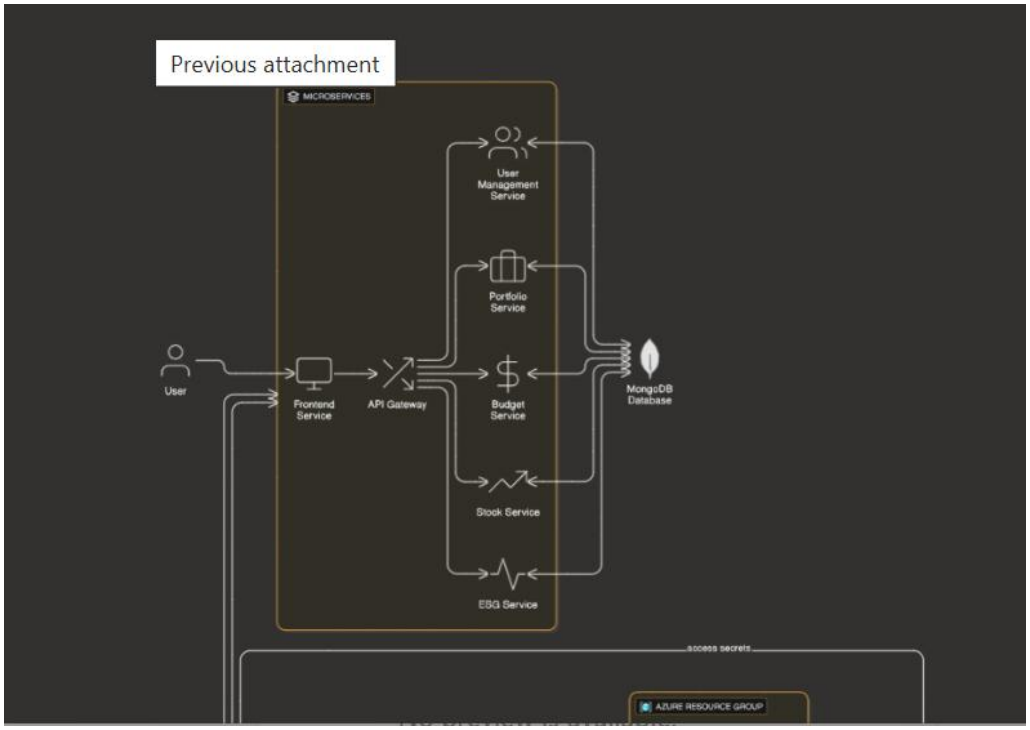
d. **User assigned managed identity** for accessing key vaults..using by aks clusters
   1. Aks clusters codes are of very large and its included in the repository I will give last..

```
# ==========================================================================================
# User Assigned Managed Identity (Used by AKS to access Key Vaults)
# This identity needs to be created BEFORE AKS clusters are configured to use it.
# ==========================================================================================
resource "azurerm_user_assigned_identity" "aks_kv_identity" {
  resource_group_name = azurerm_resource_group.myrg.name
  location            = var.location # Can stay in the primary location or a common one
  name                = "abhi-aks-kv-identity"
}
```

e. Access policy of keyvault

```
resource "azurerm_key_vault_access_policy" "kv_access_for_terraform_executor_centralindia" {
  key_vault_id = azurerm_key_vault.kv_centralindia.id
  tenant_id    = data.azurerm_client_config.current.tenant_id

  # THIS IS YOUR OBJECT ID FROM THE PREVIOUS ERROR MESSAGE
  # 'oid=3cb026dd-3f46-470c-b65c-e0bd92d4fd19'
  object_id    = "30c745d3-3b09-46f4-b0d6-b7eb7a95148a"

  # Permissions needed for Terraform to 'Get', 'List', 'Set' (create/update), and 'Delete' secrets
  secret_permissions = ["Get", "List", "Set", "Delete"]

  depends_on = [
    azurerm_key_vault.kv_centralindia
  ]
}
```

12. Infrastructure diagram:



13. How my infrastructure works:

a. I have created 2 aks clusters of 2 different region
b. I have created a key vault
c. I created the managed identity of the key vault to access the secrets
d. The secrets are:
    - username,password,and the connection string of my mongodb database
e. so the secrets should be in aks then only the pods coming in the aks can use it and be in running state
f. for that I have added the secret store csi driver in my terraform code,allow our aks pods to mount secrets from key vaults directly into their filesysytems as volumes ,which simplifies secret access for application- secured way

```
esource "azurerm_kubernetes_cluster" "aks" {
  name                = "abhiaks1"
  location            = var.vnet1_location
  resource_group_name = var.resource_group_name
  dns_prefix          = "abhiaks1"
  key_vault_secrets_provider {
    secret_rotation_enabled  = true
    secret_rotation_interval = "2m"
  }
```

g. I have written yaml files of secretproviderclass for both regions
h. I deployed the files in both regions
i. **Secretproviderclass**

secretproviderclass-centralindia.yaml

Contents History Compare Blame

```yaml
apiVersion: secrets-store.csi.x-k8s.io/v1
kind: SecretProviderClass
metadata:
  name: kv-secrets
  namespace: default
spec:
  provider: azure
  parameters:
    usePodIdentity: "false"
    useVMManagedIdentity: "true"
    clientID: "26c8778b-c389-4a71-b72f-202abd015250"
    tenantId: "ec6e8fb5-4284-4351-b9af-a68b18acee8e"
    keyvaultName: "abhi-kv-central-india"
    cloud: "AzurePublicCloud"
    objects: |
      array:
        - |
          objectName: DbUsername
          objectType: secret
          objectVersion: ""
        - |
          objectName: DbPassword
          objectType: secret
          objectVersion: ""
        - |
          objectName: mongo-uri
          objectType: secret
          objectVersion: ""
  secretObjects:
    - secretName: db-secret
      type: Opaque
      data:
        - key: dbuser
```


```yaml
        - |
          objectName: DbPassword
          objectType: secret
          objectVersion: ""
        - |
          objectName: mongo-uri
          objectType: secret
          objectVersion: ""
  secretObjects:
    - secretName: db-secret
      type: Opaque
      data:
        - key: dbuser
```


```yaml
          objectName: DbUsername
        - key: dbpassword
          objectName: DbPassword
    - secretName: mongo-secret
      type: Opaque
      data:
        - key: mongo-uri
          objectName: mongo-uri
```

```
devuser [ ~ ]$ kubectl get secretproviderclass --all-namespaces
NAMESPACE   NAME          AGE
default     kv-secrets    3d
devuser [ ~ ]$
```

it provides the instruction for the csi driver to access which key vault to connect and retrive the secrets so the pod can access the secret through the secret provider class

14. How pods are getting this secretprovider class:

a. I have mentioned in the env of deployment file of every microservice:

```yaml
      app: userservice
    annotations:
      secrets-store.csi.k8s.io/pod-sync-content: "true"
  spec:
    containers:
      - name: userservice
        image: abhiacr1.azurecr.io/userservice
        imagePullPolicy: Always
        ports:
          - containerPort: 9003
        env:
          - name: SPRING_DATA_MONGODB_URI
            valueFrom:
              secretKeyRef:
                name: mongo-secret
                key: mongo-uri
        volumeMounts:
          - name: secrets-store-inline
            mountPath: /mnt/secrets-store
            readOnly: true
    volumes:
      - name: secrets-store-inline
        csi:
          driver: secrets-store.csi.k8s.io
          readOnly: true
          volumeAttributes:
            secretProviderClass: kv-secrets
            resourceID: "/subscriptions/1674f375-e996-4423-bd25-e0e8f6e76d13/resourceGroups/abhi-resource-group/providers/Microsoft.Ma
```

b. Kubectl get pods

```
devuser [ ~ ]$ kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
apigateway-6b64d44cb4-4ggq2           1/1     Running   0          16h
azurecapstoneproject-5c9f5987d9-8t4rc 1/1     Running   0          16h
budgetservice-fd74b7744-zz6v9         1/1     Running   0          16h
esgservice-55c6cc8f74-888kq           1/1     Running   0          16h
frontend-798f9bbf5d-45gps             1/1     Running   0          15h
portfolioservice-6c9b7cd88f-t8vdl     1/1     Running   0          16h
serviceregistry-575df6df5b-ztdlg      1/1     Running   0          16h
stockservice-654746597f-xp4xj         1/1     Running   0          16h
userservice-66d7b6f44b-lzvf5          1/1     Running   0          16h
devuser [ ~ ]$
```

c. If we check logs we can see **the backend is connected to database** or not

15. Connection b/w backend and database:

a. Changed the application.properties of every microservice:

/ Files / user-service

master / user / src / main / resources /

application.properties

Contents History Compare Blame

```
1 logging.level.org.springframework=DEBUG
2 spring.data.mongodb.uri=${SPRING_DATA_MONGODB_URI}
3 spring.data.mongodb.database=abhi-db-cluster
4
5
```

b. Also changed the application.yml of every microservice :

1. Instead of localhost used external ip of eureka service

ii. if backend is connected to database...we can see logs without error

a. eg: logs of user-service

```
urekaApplications]
2025-06-25 08:01:58.561 DEBUG 1 --- [tbeatExecutor-0] o.s.web.client.RestTemplate              : HTTP PUT http://abhieureka.trafficmanager.net:8761/eureka/a
pps/USERS-SERVICE/userservice-66d7b6f44b-lzvf5:USERS-SERVICE:9003?status=UP&lastDirtyTimestamp=1750777067483
2025-06-25 08:01:58.561 DEBUG 1 --- [tbeatExecutor-0] o.s.web.client.RestTemplate              : Accept=[application/json, application/*+json]
2025-06-25 08:01:58.566 DEBUG 1 --- [tbeatExecutor-0] o.s.web.client.RestTemplate              : Response 200 OK
2025-06-25 08:01:59.192 DEBUG 1 --- [freshExecutor-0] o.s.web.client.RestTemplate              : HTTP GET http://abhieureka.trafficmanager.net:8761/eureka/a
pps/delta
2025-06-25 08:01:59.192 DEBUG 1 --- [freshExecutor-0] o.s.web.client.RestTemplate              : Accept=[application/json, application/*+json]
2025-06-25 08:01:59.196 DEBUG 1 --- [freshExecutor-0] o.s.web.client.RestTemplate              : Response 200 OK
2025-06-25 08:01:59.196 DEBUG 1 --- [freshExecutor-0] o.s.web.client.RestTemplate              : Reading to [org.springframework.cloud.netflix.eureka.http.E
urekaApplications]
devuser [ ~ ]$
```

c. like this from every pods logs we can confirm that backend is properly connected to database or not

d. everything is properly working:

```
NAME                                    READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/apigateway              1/1     1            1           43h
deployment.apps/azurecapstoneproject    1/1     1            1           5d21h
deployment.apps/budgetservice           1/1     1            1           23h
deployment.apps/esgservice              1/1     1            1           42h
deployment.apps/frontend                1/1     1            1           43h
deployment.apps/portfolioservice        1/1     1            1           25h
deployment.apps/serviceregistry         1/1     1            1           5d
deployment.apps/stockservice            1/1     1            1           3d23h
deployment.apps/userservice             1/1     1            1           2d12h

NAME                        TYPE           CLUSTER-IP     EXTERNAL-IP        PORT(S)          AGE
service/apigateway          LoadBalancer   10.0.212.58    135.235.188.243    9000:31528/TCP   43h
service/budgetservice       LoadBalancer   10.0.76.3      98.70.236.117      9002:30274/TCP   23h
service/esgservice          LoadBalancer   10.0.84.121    4.188.191.156      9004:31698/TCP   42h
service/frontend-service    LoadBalancer   10.0.59.73     20.235.200.171     80:31063/TCP     43h
service/kubernetes          ClusterIP      10.0.0.1       <none>             443/TCP          6d19h
service/portfolioservice    LoadBalancer   10.0.56.19     20.235.215.27      9001:31043/TCP   2d1h
service/serviceregistry     LoadBalancer   10.0.124.230   52.140.85.66       8761:31279/TCP   5d
service/stockservice        LoadBalancer   10.0.44.73     20.235.234.190     8081:32137/TCP   3d23h
service/userservice         LoadBalancer   10.0.143.210   20.235.214.89      9003:32258/TCP   5d2h

NAME                                      READY   STATUS    RESTARTS   AGE
pod/apigateway-6b64d44cb4-4gqq2           1/1     Running   0          16h
pod/azurecapstoneproject-5c9f5987d9-8t4rc 1/1     Running   0          17h
pod/budgetservice-fd74b7744-zz6v9         1/1     Running   0          17h
pod/esgservice-55c6cc8f74-888kq           1/1     Running   0          17h
pod/frontend-798f9bbf5d-45gps             1/1     Running   0          15h
pod/portfolioservice-6c9b7cd88f-t8vdl     1/1     Running   0          17h
pod/serviceregistry-575df6df5b-ztdlg      1/1     Running   0          17h
pod/stockservice-654746597f-xp4xj         1/1     Running   0          17h
pod/userservice-66d7b6f44b-lzvf5          1/1     Running   0          17h
devuser [ ~ ]$
```

16. Connect frontend to backend and check the connections are working properly or not:

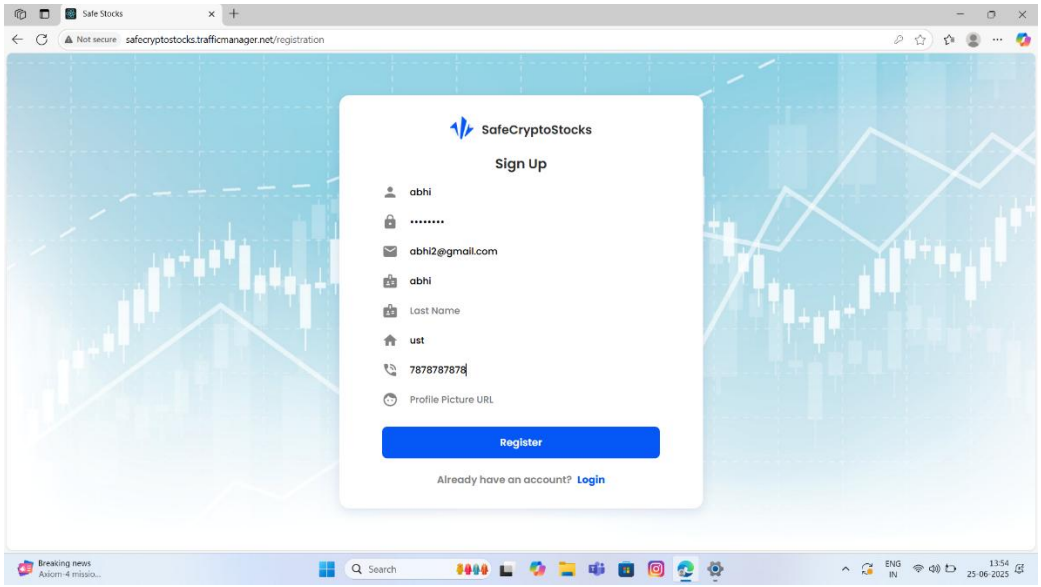
a. Everywhere in frontend I changed the localhost into external ip of load balancer of api gateway
b. The issues of cors error also get fixed
c. From frontend the traffic is routed to api gateway..the api gateway transfer to diff microservices
d. Everything is working properly

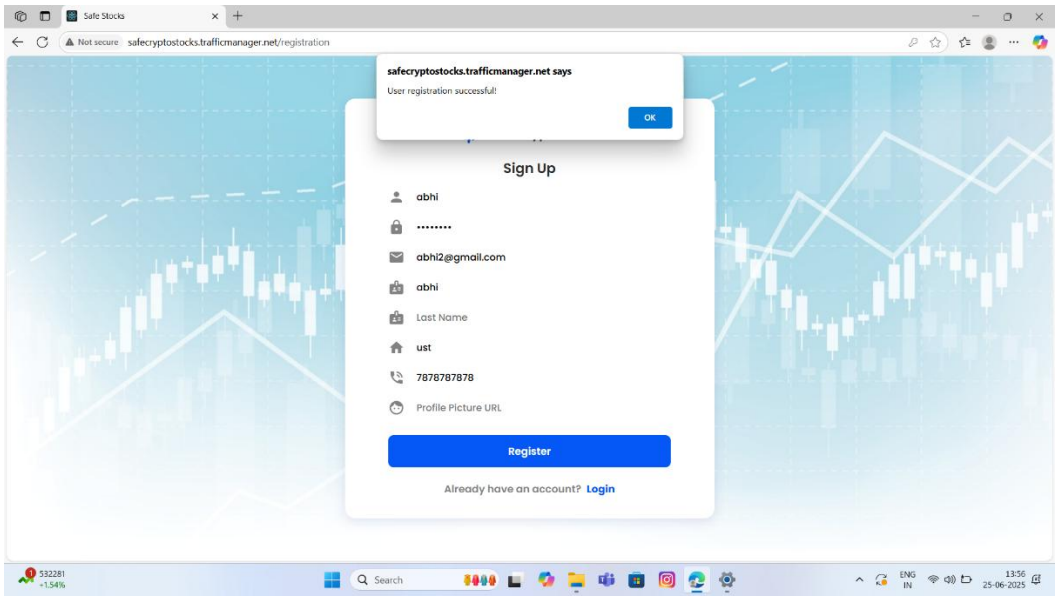
17. Tried accessing the application using traffic manager:

a. Created a **traffic manager** for each service and added end points as external ip of both regions of particular services.
b. changed the external ip **into traffic manager dns in every configurations**

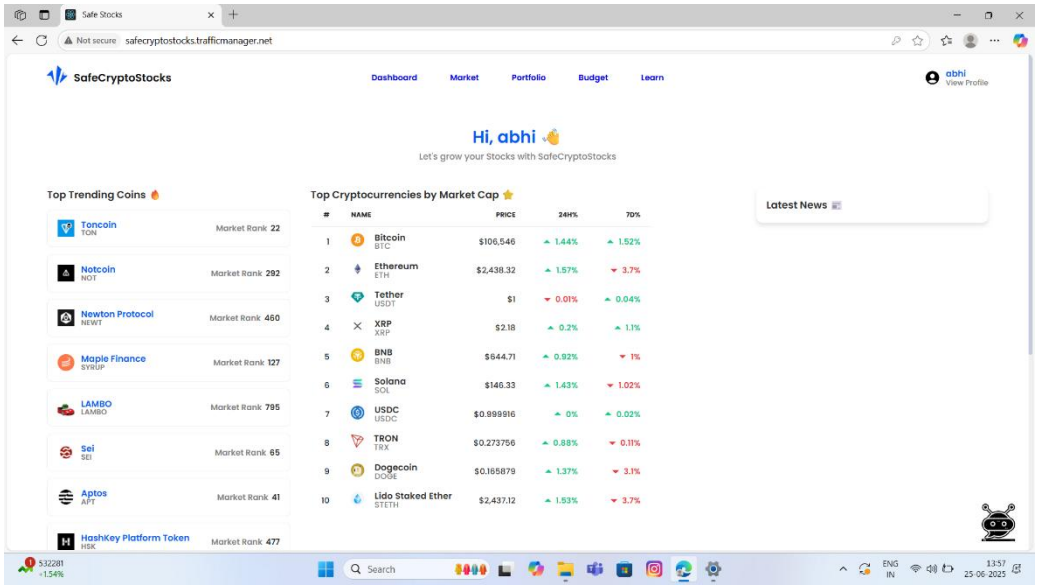
18. Accessing application:

a. Register:

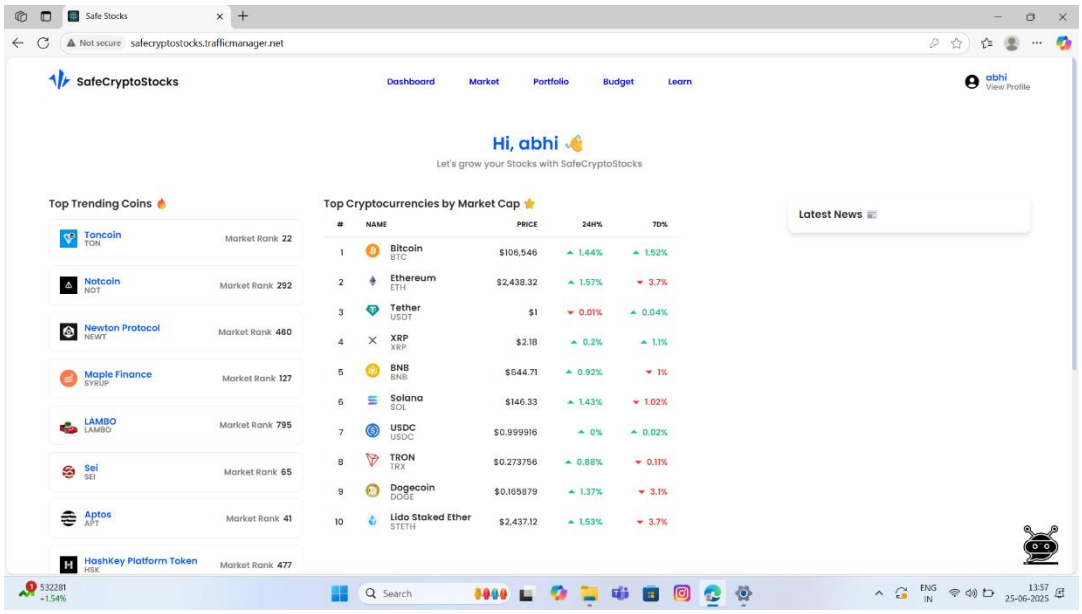


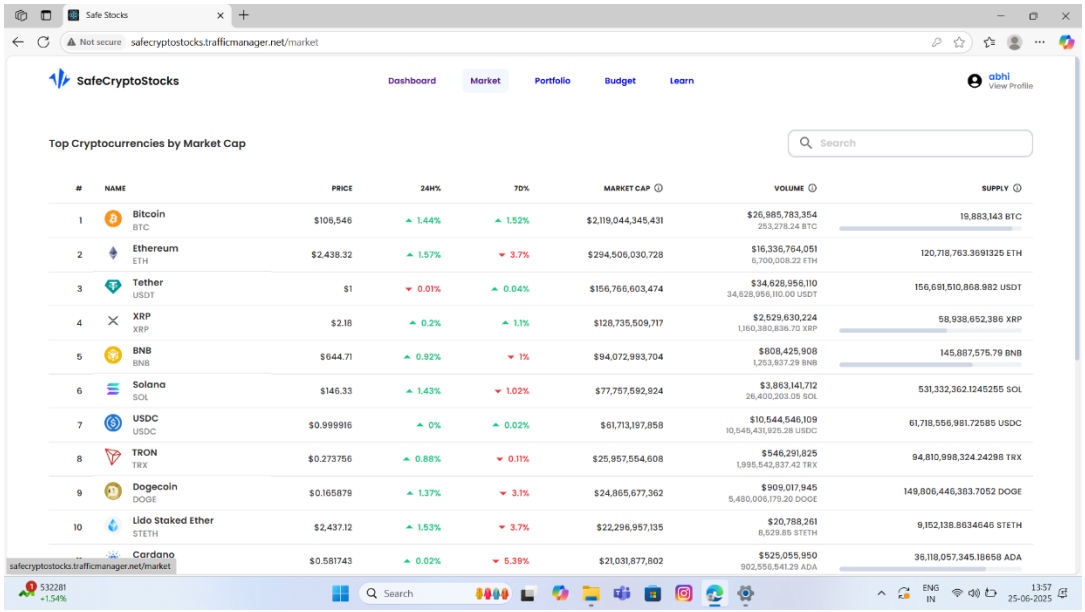


i. Login:

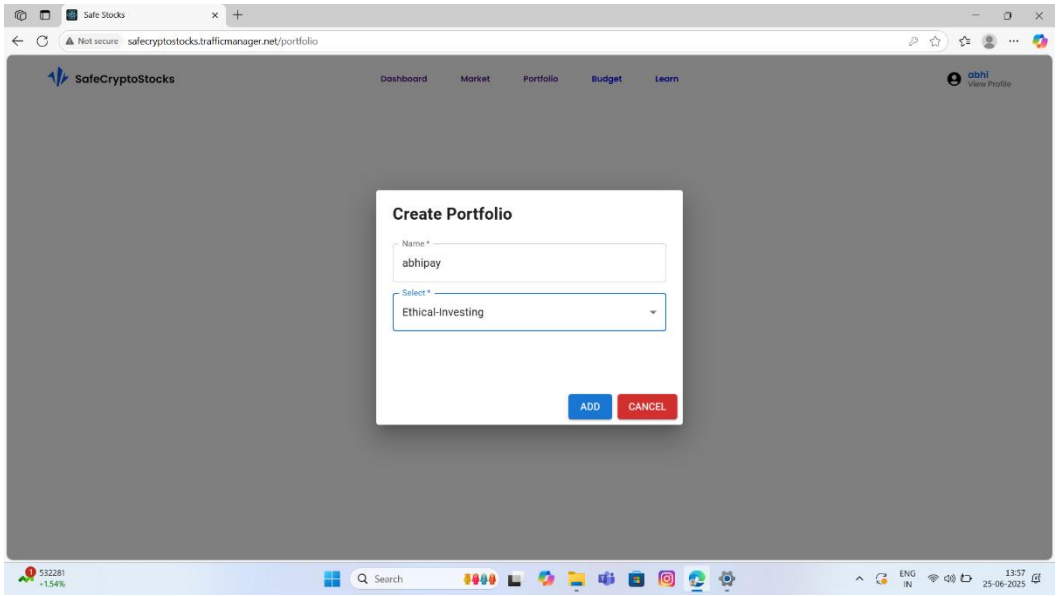


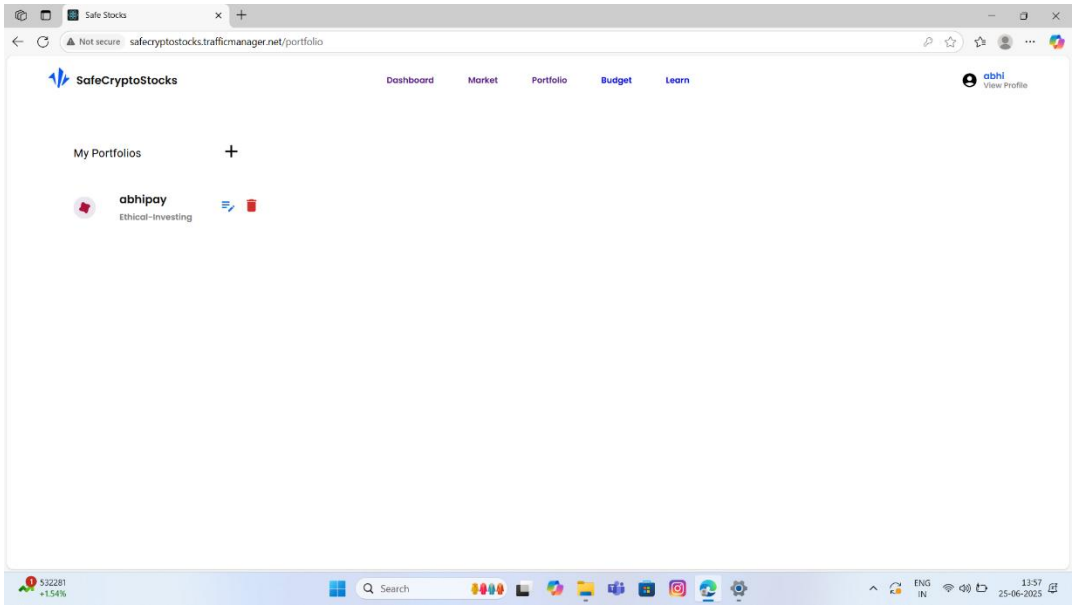
ii. Stock:



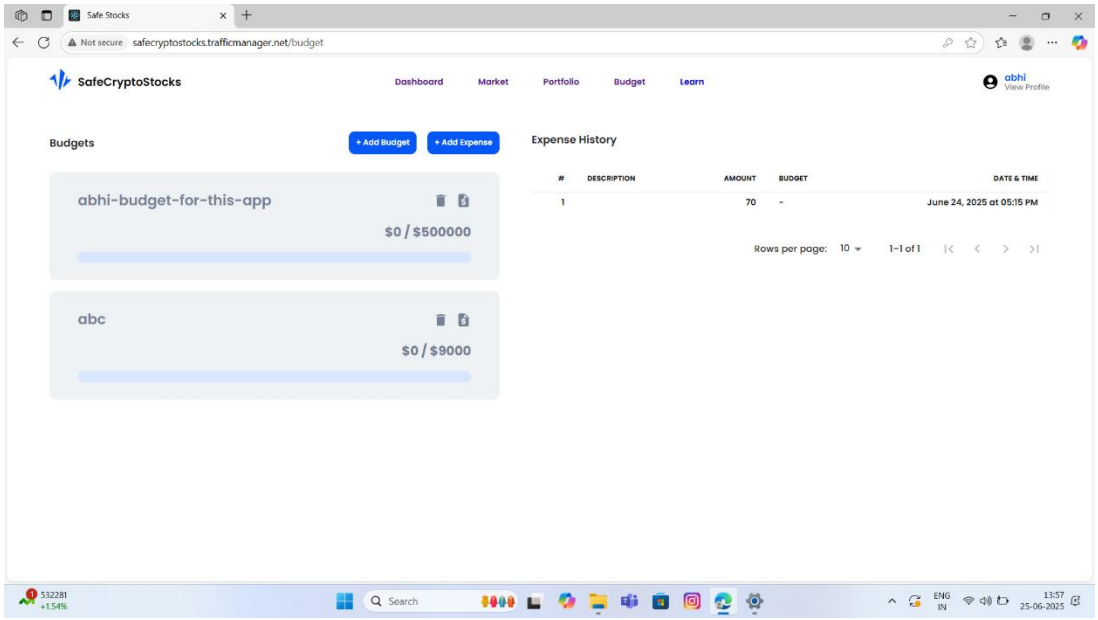


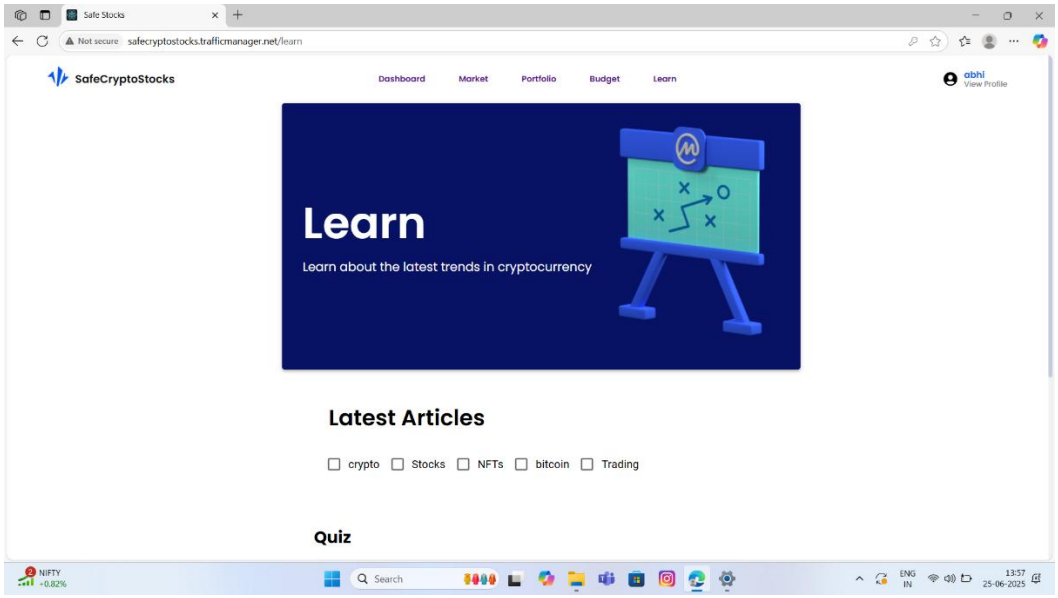
iii. Portfolio:



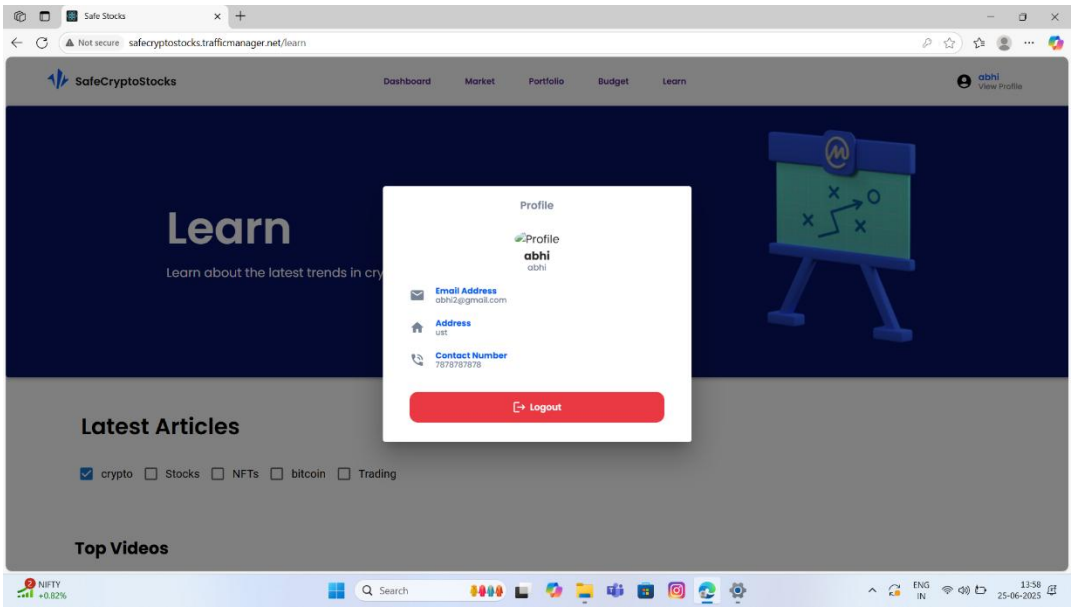

Safe Stocks
Not secure safecryptostocks.trafficmanager.net/portfolio
SafeCryptoStocks
Dashboard
Market
Portfolio
Budget
Learn
abhi
View Profile
My Portfolios
abhipay
Ethical-Investing

iv. Budget:


Safe Stocks
Not secure safecryptostocks.trafficmanager.net/budget
SafeCryptoStocks
Dashboard
Market
Portfolio
Budget
Learn
abhi
View Profile
Budgets
+ Add Budget
+ Add Expense
abhi-budget-for-this-app
$0 / $500000
abc
$0 / $9000
Expense History
DESCRIPTION AMOUNT BUDGET DATE & TIME
1 70 - June 24, 2025 at 05:15 PM
Rows per page: 10
1–1 of 1



Logout:



19. NSG associated with the vm:

a. Frontend is hitting on 80

Search rules | Source == all | Destination == all | Protocol == all | Action == all

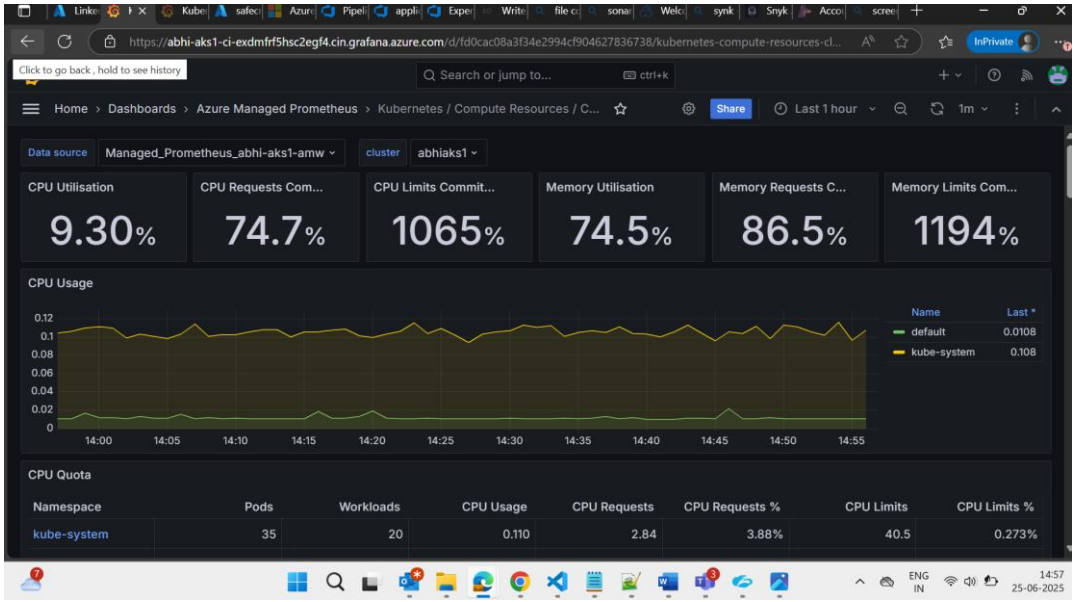
Priority ↑	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (7)						
300	SSH	22	TCP	Any	Any	Allow
309	AllowAnyCustom80Inbound	80	TCP	Any	Any	Allow

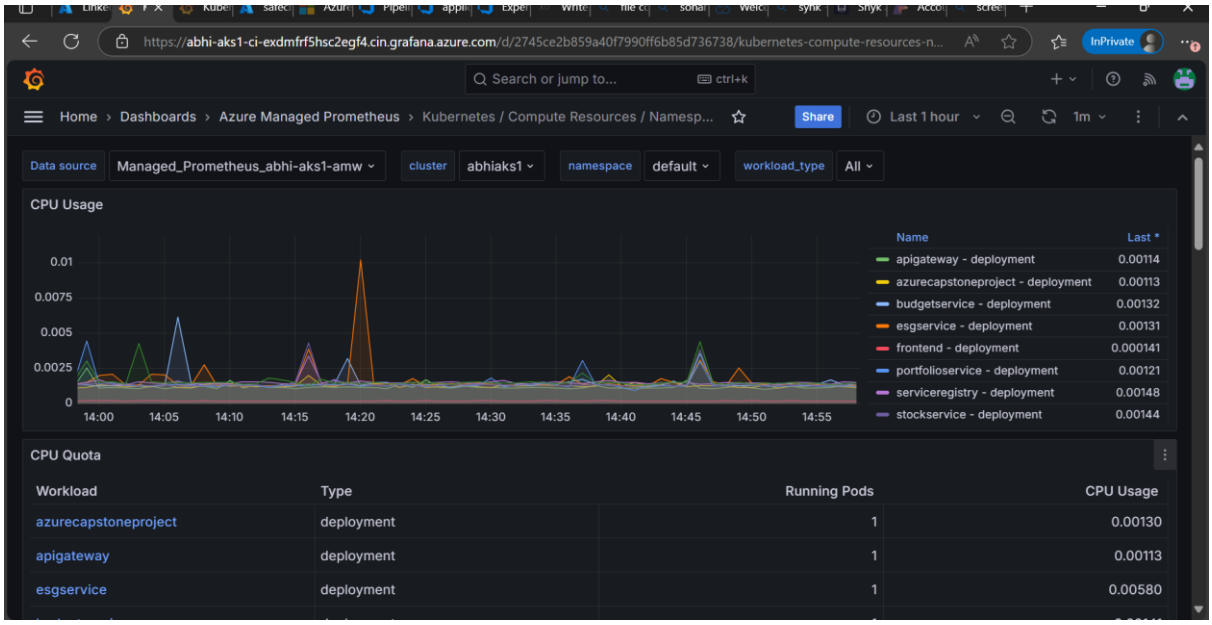
20. Grafana

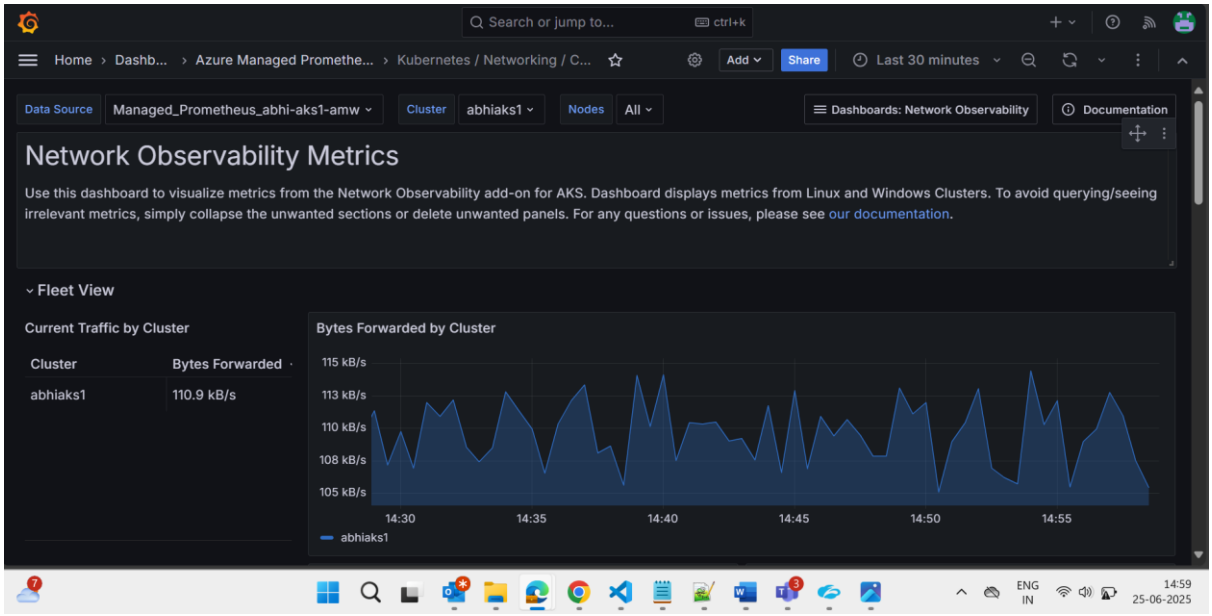
a. Configured Grafana for aks1

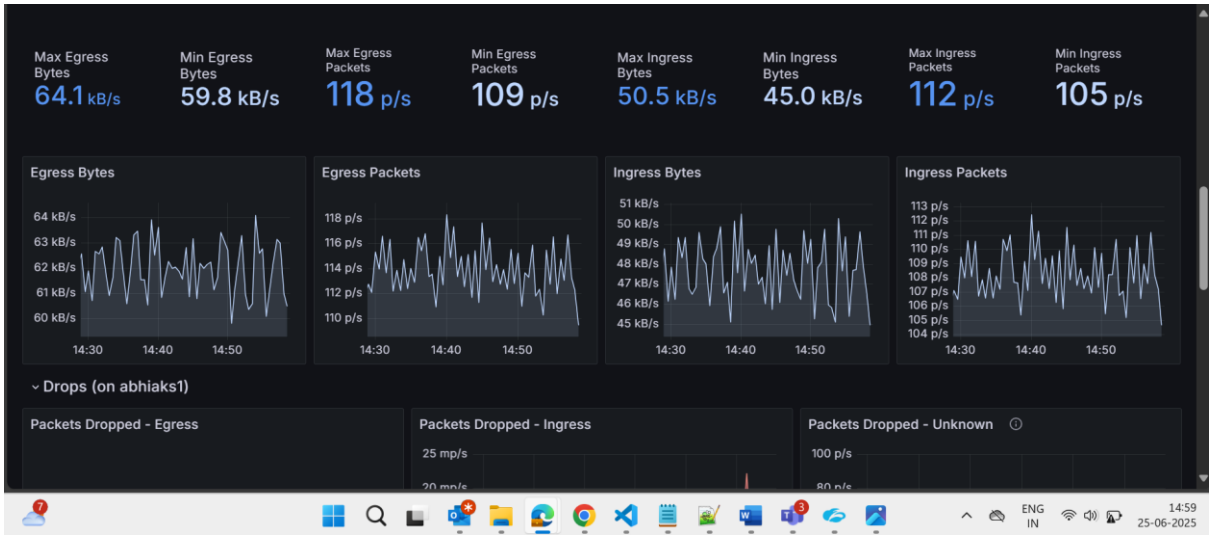
b. Steps:

1. Azure Managed Grafana
2. Azure Monitor workspaces (configure Grafana with workspace)
3. Kubernetes services (configure workspace with k8s service)
4. Grafana





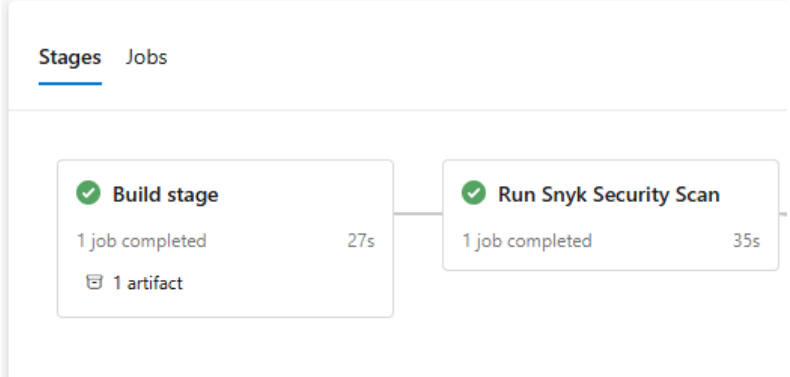




21. Snyk:

a. Tried snyk in the pipeline
b. Synk token got from [Account general settings | Snyk]
c. Change the pipeline configuration before deployment

```
# ==================== Snyk Security Scan ====================
# - stage: SecurityScan
#   displayName: Run Snyk Security Scan
#   dependsOn: Build
#   jobs:
#   - job: SecurityScan
#     displayName: Run Snyk for SCM
#     pool:
#       name: abhipool
#       vmImage: $(vmImageName)
#     steps:
#     - task: NodeTool@0
#       inputs:
#         versionSpec: '20.x'
#       displayName: 'Ensure Node.js for Snyk'

#     - script: |
#         echo "Installing Snyk CLI..."
#         npm install -g snyk
#       displayName: 'Install Snyk CLI'

#     - script: |
#         echo "Authenticating with Snyk..."
#         snyk auth $(SNYK_TOKEN)
#       


#     - script: |
#         snyk test --all-projects --severity-threshold=medium --json > snyk-report.json
#         cat snyk-report.json
#       displayName: 'Run Snyk Test'
```



22. Challenges

a. Multiple User Assigned Identities Error
b. ImagePullBackOff / 401 Unauthorized Error
c. Control Plane Deallocated
d. Snyk I tried and the deployment was very slower so I removed snyk

How I resolved it?

- Multiple user assigned identities error:
  - Ensured only 1 user assigned identity in secretproviderclass
  - Removed conflicting identities from aks node pools
- Image pull backoff error:
  - Forgot to give a annotations in deployment.yaml file
- control plane deallocated:
  - iiht issue
  - resolved by updating the cluster
  - **az aks update --name abhi-aks2 --resource-group abhi-resource-group**
  - done this on both the clusters

23. **pushed each service into github:**

i. **frontend:** https://github.com/Abhiramikannan/safecryptostocks-frontend.git
ii. **api gateway:** https://github.com/Abhiramikannan/safecryptostocks-apigateway.git
iii. **eureka service:** https://github.com/Abhiramikannan/safecryptostocks-eureka-service.git
iv. **user-service**: https://github.com/Abhiramikannan/safecryptostocks-user.git
v. **stock-service**: https://github.com/Abhiramikannan/safecryptostocks-stock.git
vi. **portfolio-service:** https://github.com/Abhiramikannan/safecryptostocks-portfolio.git
vii. **esg-service:** https://github.com/Abhiramikannan/safecryptostocks-esg.git
viii. **terraform code:** https://github.com/Abhiramikannan/safecryptostocks-terraform.git